A constructive neutral memory kernel

Ordered Wick descendants, energy regression, and exact static removal

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7 November 2026 Local research draft

Abstract

We extend the preceding rank-marked chord rule to all ordered Gaussian-Wick descendants in the neutral memory field. Internal field pairings give an exact rule for Gaussian-chaos projections. A cut-diagram argument identifies deterministic energy regression and cancels all diagrams without a bridge between the two fields. Every surviving diagram has exponential suppression in the external Hermite degrees. Consequently each fixed descendant, after regression, has an absolutely continuous two-energy spectrum. This permits the limit of exact samplewise energy pinching. The previously proved uniform time-series tails then give a convergent construction of the full neutral finite-time covariance. The charged defect and its cross covariance remain to be computed.

1 Setting and the remaining neutral problem

Use standard CAR, overlapping complex Gaussian p-body couplings, the same original couplings in all charge sectors, and the normalized annealed full trace at $\beta = \mu = 0$. Take even N, $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$, fixed J_s, $\hbar = 1$, and string time. Set $C = J_s^2/\lambda$, $q = e^{-4\lambda} \in (0, 1)$ and

$$Z_j = \mathcal{W}_j(H^j), \quad U_j = (jp - \mathcal{R})Z_j, \quad P_j(q) = [j]_q!, \quad \mathcal{D}_j = j^2 - 4q\partial_q. \tag{1}$$

Here $\mathcal{W}_j$ is ordinary coupling-Gaussian chaos projection. All polynomial degrees are fixed before the double-scale limit. The previous stage [3] replaces U_j strongly by $(jp - \mathcal{L})Z_j$, where $\mathcal{L}$ is half the Clifford length, and gives their two-mark chord moments. The preceding series [4] reduces the neutral memory to these fields and forced lower-chaos descendants, with uniform summable tails. We now compute those descendants and their static part. No independent-probe substitution is made.

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2 Primitive descendants and the Gaussian-chaos label

Write

$$T_{v,w} = U_v \diamond Z_w, \qquad F_{r,v,w} = H^r T_{v,w}, \quad v \geq 1,\ w, r \geq 0. \tag{2}$$

The diamond is the ordinary Gaussian Wick product with ordered matrix coefficients. Its entire $v + w$ source block forbids internal Gaussian pairings; the physical size mark acts only on its first v slots. The r ordinary slots may contract among themselves or with the source. Taking an adjoint reverses all slots, so a prefix mark becomes a suffix mark. These two types of block must be stored separately.

For two primitive fields separated by H^a, H^b, sum over complete pairings π obeying the source-block restrictions. If $L_F = r + v + w$ and $i_F(\pi)$ pairs have both ends in the whole field interval, put

$$d_F(\pi) = L_F - 2i_F(\pi). \tag{3}$$

This is the degree of the field's uncontracted Gaussian Wick tensor. An insertion of $\mathcal{W}_d F$ keeps only $d_F = d$; an insertion of $\mathsf{O}_q F = q^{\mathcal{N}} F$ multiplies the diagram by q^{d_F}. External energy slots are never counted in L_F.

Let I_1, I_2 be the chord sets incident to the two marked subblocks, and use the preceding geometrical weight

$$K(I) = |I|^2 - 4\,\mathrm{cr}(\pi|_I), \quad K_1 = K(I_1),\ K_2 = K(I_2),\ K_{12} = K(I_1 \cap I_2). \tag{4}$$

The zero-, one-, and two-mark coefficients are respectively

$$1, \qquad \lambda K_1, \qquad \lambda K_{12} + \lambda^2 K_1 K_2, \tag{5}$$

all multiplied by $C^{L/2} q^{\mathrm{cr}(\pi)}$, where L is the total number of raw Hamiltonian slots. Explicit scalar factors of C in a descendant recurrence are retained in addition to this weight. Odd L gives zero. This is an algorithm for every fixed degree, including chaos projections; it is not an assumption of spectral q-Wick ordering at finite N.

The extension from rank to local size remains strong: right Wick creation satisfies

$$\|X_v \diamond Z_w\|_2 \leq \sqrt{(v+w)!/v!}\, M^{w/2} \|X_v\|_2, \qquad \sup_m C_m \leq M. \tag{6}$$

Fixed ordinary powers of H are bounded on fixed Gaussian degree, and chaos projections and $q^{\mathcal{N}}$ are contractions. Thus only the already proved two-field L^2 bridge is needed.

3 Energy regression from a cut field diagram

Let $H_d^{(q,C)}$ denote the monic q-Hermite polynomial,

$$H_{d+1}^{(q,C)}(x) = xH_d^{(q,C)}(x) - C[d]_q H_{d-1}^{(q,C)}(x), \qquad \|H_d^{(q,C)}\|^2_{L^2(\rho_q)} = C^d P_d(q), \tag{7}$$

where ρ_q is the energy law of variance C. Define $\operatorname{Reg} F$ by its limiting inner products with energy polynomials. For a pure source one obtains

$$\boxed{\operatorname{Reg} T_{v,w} = \kappa_v H_{v+w}^{(q,C)}, \qquad \kappa_v = \lambda \frac{\mathcal{D}_v P_v(q)}{P_v(q)}.} \tag{8}$$

Indeed, pair the source with a Wick block of length $v + w$. All chords are bridges. Their permutations have weight $q^{\operatorname{inv}\sigma}$. The Mahonian decomposition into the first v positions, the remaining w positions, and their shuffle shows that the first group's internal permutation has its own q^{inv} law. Its conditional average of K_1 is $\mathcal{D}_v P_v / P_v$, independent of the selected endpoint set. The suffix case follows by the same decomposition.

There is a finite formula for every projected primitive. Fix a partial matching γ internal to $F_{r,v,w}$, respecting the source restriction. Its remaining $d = L_F - 2|\gamma|$ slots are Gaussian Wick ordered. Cut open the field interval and number all their external partners in circular order. Crossings of internal chords with one another and with open legs give a number c_γ independent of the external assignment. Crossings between open legs have count $\operatorname{inv}\sigma$ for a suitable order convention. The mark has a value $K_\gamma(\sigma)$ depending only on this finite configuration. Thus its regression coefficient is

$$g_{\gamma,d} = C^{|\gamma|} q^{c_\gamma} \lambda \frac{\sum_{\sigma \in S_d} q^{\operatorname{inv}\sigma} K_\gamma(\sigma)}{P_d(q)}. \tag{9}$$

Summing over γ and the field's scalar coefficients gives $\operatorname{Reg}(\mathcal{W}_d F) = g_{F,d} H_d^{(q,C)}$. The case $d = 0$ is included. Formula (3) shows that $q^{\mathcal{N}}$ simply multiplies this group by q^d.

This derivation can also be checked by pairing with one external $H_d^{(q,C)}$: the same permutation sum is exactly $\langle F, H_d^{(q,C)}(H)\rangle / (C^d P_d)$. The leading $Z_d - H_d^{(q,C)}(H) \to 0$ bridge is used only in such linear inner products. It is not used to infer continuity of arbitrary products of marked fields.

4 No-bridge cancellation and the external Hermite bound

Consider two projected primitives after subtracting their energy regressions. Fix their internal matchings γ, γ'. First sum diagrams having no chord directly joining the two field intervals. Then $I_1 \cap I_2 = \varnothing$, so $K_{12} = 0$. With all external partner positions fixed, the two open-endpoint permutations are independent. Reordering endpoints within one field does not change crossings with any chord having both endpoints outside that field. Therefore the two one-mark permutation sums factor exactly as (9).

Subtracting regression cancels this entire no-bridge group, including all internal partial matchings. The statement remains true with $q^{\mathcal{N}}$ or a fixed chaos projection, whose coefficients are constant on each internal-matching group. An individual no-bridge diagram need not vanish.

Proposition 1. *Let $R_F = F - \operatorname{Reg} F(H)$ and $R_G = G - \operatorname{Reg} G(H)$, where F, G are finite combinations of the primitives, their chaos projections, and $q^{\mathcal{N}}$ images. Their limiting two-energy measure has an $L^2(\rho_q \otimes \rho_q)$ density. In particular it has no mass on the diagonal $E = E'$.*

Proof. Use the normalized external basis $p_n = H_n^{(q,C)}/\sqrt{C^n P_n}$ and set

$$M_{mn} = \lim \mathbb{E} \operatorname{tr}[R_F^\dagger p_m(H) R_G p_n(H)]. \tag{10}$$

The external q-Hermite polynomials implement the standard multiblock Wick cancellations [1]: no pair remains internal to either energy block. A deleted energy-internal chord belongs to neither marked incident set; deleting it changes no crossing between the remaining chords. Thus the marked weight is preserved as a coefficient in this cancellation. This is a diagram identity.

Let D bound the sum of the two raw field lengths. After the no-bridge cancellation, there is at least one field-to-field bridge. If r chords connect the two energy blocks, endpoint counting gives

$$|m - n| \le D, \qquad r \ge \min(m, n) - D. \tag{11}$$

Every field bridge crosses every energy bridge, hence contributes at least a factor q^r. Choosing the finite field attachments costs at most a constant times $(m + n + 1)^D$. The remaining energy bridge permutations have total weight $P_r(q)$. Since

$$\frac{P_r(q)}{\sqrt{P_m(q) P_n(q)}} \le (q; q)_\infty^{-1}, \qquad r \le \min(m, n), \tag{12}$$

the finite mark and regression coefficients give

$$\boxed{|M_{mn}| \le K_D (m + n + 1)^D q^{\max(0, \min(m,n) - D)} \, 1_{|m-n| \le D}.} \tag{13}$$

Here K_D depends on the fixed fields, C, λ, q, never on m, n. In particular $\sum_{m,n} |M_{mn}|^2 < \infty$. □

5 From the bound to the spectrum and exact pinching

For a finite field list define the positive matrix measures by

$$\nu_{jk,N}(A,B) = \mathbb{E}\,\mathrm{tr}[R_{j,N}^{\dagger} P_H(B) R_{k,N} P_H(A)]. \tag{14}$$

Their fixed monomial moments follow from the finite diagram rule. Absolute marked weights are bounded by fixed constants times ordinary q-Gaussian moments. Positivity, higher-moment uniform integrability and spectral Cauchy–Schwarz give weak convergence to a unique measure supported on

$$[-R_H, R_H]^2, \qquad R_H = 2\sqrt{C/(1-q)}. \tag{15}$$

As in the preceding stage, this concerns fixed field degrees and all fixed external degrees; no uniform finite-N estimate in the external degree is asserted.

The bound (13) gives the density constructively:

$$f_{jk}(E,E') = \sum_{m,n\geq 0} M_{mn}^{jk} p_n(E) p_m(E'), \qquad d\nu_{jk} = f_{jk}\, d\rho_q(E) d\rho_q(E'). \tag{16}$$

The series converges in L^2. Its polynomial moments agree with the original measure, and polynomials are dense on the compact support, proving equality of measures. The matrix density is positive almost everywhere. This uses the Hilbert–Schmidt criterion also underlying [2], not an unproved identification of these common-coupling fields with independent q-Fock probes. The diffuse measure $\rho_q \otimes \rho_q$ assigns zero mass to $\Delta = \{E = E'\}$.

For a residual field with this limit, exact pinching satisfies

$$\|\mathcal{P}_H R_N\|_2^2 = \nu_{RR,N}(\Delta), \qquad \mathcal{P}_H X = \sum_e P_e X P_e. \tag{17}$$

The diagonal is closed. Weak convergence of the finite positive measures and Portmanteau therefore imply

$$\limsup_N \|\mathcal{P}_H R_N\|_2^2 \leq \nu_{RR}(\Delta) = 0. \tag{18}$$

This retains entire degenerate energy blocks and proves annealed L^2 convergence; it is not an interchange of an infinite-time average with the double-scale limit.

An explicit coefficient-tail bound follows from (13):

$$\sum_{\min(m,n)\geq L} |M_{mn}|^2 \leq 2(2D+1) K_D^2 \sum_{n\geq L} (2n+D+1)^{2D} q^{2(n-D)} \quad (L \geq D). \tag{19}$$

The scalar series has a geometric upper bound once its successive ratio is below one. Also $(q;q)_\infty^{-1} \leq \exp[q/(1-q)^2]$ supplies a finite explicit normalization bound.

6 The forced descendants and their full static polynomials

Suppress finite-N subscripts after taking each fixed-degree limit. The source and descendant recurrence are

$$r_j = 4\lambda C \sum_{v<j} q^v T_{v,j-1-v}, \tag{20}$$

$$\eta_{j+1} = H\eta_j - C[j]_q \eta_{j-1} - r_j, \qquad \eta_0 = \eta_1 = \eta_2 = 0, \tag{21}$$

with $T_{0,w} = 0$. Each η_j is a finite linear combination of $H^r T_{v,w}$, so all its diagrams are covered above. If $g_j = \operatorname{Reg} \eta_j$, energy regression commutes with multiplying by H, giving the scalar recurrence

$$\boxed{g_{j+1}(x) = x g_j(x) - C[j]_q g_{j-1}(x) - 4\lambda C \left(\sum_{v<j} q^v \kappa_v \right) H_{j-1}^{(q,C)}(x).} \tag{22}$$

Again $g_0 = g_1 = g_2 = 0$. Write $g_j = \sum_{d \le j-2} g_{j,d} H_d^{(q,C)}$. Self-adjointness of Gaussian chaos projection, or the cut rule, gives

$$\operatorname{Reg}(\mathsf{O}_q \eta_j) = \sum_d q^d g_{j,d} H_d^{(q,C)}. \tag{23}$$

The unpinched coefficients of the neutral series are

$$V_j = 4\lambda q^j U_j + (q^j - \mathsf{O}_q)\eta_j, \tag{24}$$

$$v_j(x) := \operatorname{Reg} V_j = 4\lambda q^j \kappa_j H_j^{(q,C)}(x) + \sum_{d \le j-2} (q^j - q^d) g_{j,d} H_d^{(q,C)}(x), \tag{25}$$

$$R_j = V_j - v_j(H). \tag{26}$$

Every R_j has the absolutely continuous spectrum (16), and its exact pinching vanishes in the limit.

For calibration,

$$g_3 = -4\lambda^2 C q H_1^{(q,C)}, \tag{27}$$

$$g_4 = -\frac{8\lambda^2 C q(1+3q)}{1+q} H_2^{(q,C)} - 4\lambda^2 C^2 q. \tag{28}$$

The scalar in g_4 agrees with the previously retained chaos-zero term. No conserved term is removed before the recurrence or $q^{\mathcal{N}}$ operation. Moreover

$$R_1 = 4\lambda q(U_1 - \lambda H), \qquad d\nu_{11} = 16\lambda^3 C q^2 \, d\mu_q, \tag{29}$$

where μ_q is the normalized Mehler measure of weight q. The neutral memory slope $2CR_1$ therefore reproduces the previous force spectrum $64\lambda^3 C^3 q^2 \mu_q$.

7 The full neutral finite-time covariance

Let

$$c_j(s) = \frac{\int e^{isE} H_j^{(q,C)}(E)\, d\rho_q(E)}{C^j P_j(q)}. \tag{30}$$

These are the limiting Jacobi coefficients in [4]. That stage proved, with finite parameters retained until the limit,

$$B_{m,N}(s) = \frac{m\epsilon_m C_m}{p}(1-\mathcal{P}_h)e^{-ihs/2}\sum_{j\geq 0} c_{j,m}(\alpha_m s)V_{j,m,N}e^{-ihs/2} + o_{L^2}(1), \tag{31}$$

uniformly on compact real memory times in the full charge sum. Here $\alpha_m = (1-p/m)_+$, $\epsilon_m = 1-\alpha_m^2$; on central windows $\alpha_m \to 1$, $m\epsilon_m/p \to 2$ and $C_m \to C$. The auxiliary Jacobi entries converge and are uniformly bounded, so their operators converge strongly. Their exponentials converge strongly uniformly on compact s-intervals; this gives each $c_{j,m}(s) \to c_j(s)$. Continuity handles $\alpha_m \to 1$. The fixed-degree field coefficients converge to (25). The tails are bounded uniformly by an explicit summable envelope of the form

$$\|c_{j,m}(s)V_{j,m,N}\|_2 \leq K_S(j+1)^2 X_S^j/\sqrt{j!}. \tag{32}$$

In the limiting joint Gram space, Reg is the orthogonal projection onto the energy-polynomial closure. Hence $\|R_j\|_2^2 = \lim_N \|V_{j,N}\|_2^2 - \|v_j\|^2_{L^2(\rho_q)}$ is at most $\lim_N \|V_{j,N}\|_2^2$, so the same coefficient envelope controls the residual series. For a fixed cutoff J, subtract $v_j(H)$ and apply (18) to every coefficient. The removed polynomials commute exactly with H. Then remove J using the above tail and the contraction property of pinching. This proves the neutral covariance limit

$$\begin{aligned}&\lim_N \mathbb{E}\,\mathrm{tr}[B_N(s)^\dagger e^{iTH}B_N(u)e^{-iTH}]\\ &= 4C^2\sum_{j,k\geq 0}\overline{c_j(s)}c_k(u)\int e^{\frac{i}{2}(s-u)(E+E')+iT(E'-E)}\, d\nu_{jk}(E,E').\end{aligned} \tag{33}$$

The sum is absolutely convergent, uniformly for s,u in compact sets and every real T: spectral Cauchy–Schwarz bounds each measure's total variation by $\|R_j\|_2\|R_k\|_2$. This all-T statement concerns the series tail. The spectral $N\to\infty$ limit is for each fixed T, not uniformly on the whole real axis. All M_{mn}^{jk} in (16) are supplied by the finite diagram algorithm, so (33) is a convergent constructive answer, not a finite list of Taylor coefficients.

The spectral measures of the truncated time fields converge in total variation as $J\to\infty$. Absolute continuity is closed under this convergence, so the full neutral residual has no static spectral mass. An L^2 density for the infinite series is not required or asserted. Central uniformity and the previously proved binomial charge tails give the full-trace result. No finite-temperature, growing-time, or simultaneous $\lambda\to 0$ limit is included.

8 Verification and the next physical calculation

The checks accompanying this stage retain complete finite CAR operators for Gaussian-chaos bookkeeping and separately test the limiting chord weights. They compare finite Hermite expansions with the internal-pair rule (3); these are algebraic checks, not finite-N rank–size equalities. Independent permutation enumeration tests (8), and explicit no-bridge groups test the cancellation used in Proposition 1. The descendant driver passed 42 complete finite comparisons (maximum error $6.67 \cdot 10^{-15}$), 12 earlier-moment calibrations, 10 source regressions and 6 nonempty no-bridge cancellations. It saves 59 signed moment polynomials and 9 chaos norms; 31 negative coefficients are retained. Finite normalized CAR comparisons use floating arithmetic; formal polynomials are exact.

An independent driver enumerates 5,913 permutations through degree seven, checks 56 prefix/suffix integer-polynomial identities, and retains 12,316 negative mark values. Seven eight-slot no-bridge layouts give 168 allowed pairings and 38 partner groups, all with exact factorization. Their 120 nonzero centered single-diagram weights include 60 negative weights. This last check uses formal rational values $q = 1/3$, $\lambda = 1/4$ for polynomial identities, not a physical parameter pair satisfying $q = e^{-4\lambda}$. Both scientific drivers passed their first runs. Parameters, versions and input hashes are recorded with the outputs.

What has changed. The neutral branch now has fixed-degree joint moments for all forced descendants, their full deterministic static polynomials, an all-external-degree spectral bound, and a controlled finite-time covariance after exact energy pinching. The proof of the spectral bound uses the arbitrary-degree cut argument and bridge counting; finite enumerations alone would not establish absolute continuity.

Literature and attribution. Ordinary Gaussian Wick expansion, the multiblock q-Wick formulas of Effros–Popa [1], and the Hilbert–Schmidt strategy for generator measures in Bikram–Mukherjee [2] are established tools. Their use here requires the physical rank-to-size bridge, the common-coupling local marks, and the cut-field cancellation above. We do not infer a q-Gaussian ambient algebra for the fluctuation fields solely from the energy law being q-Gaussian.

Continuation. The actual memory field remains $\mathsf{F} = A + B + o_{L^2}(1)$, where A is the charged transfer defect. The next calculation must put its forward and reverse sources into the same common-J diagram system and determine AA, AB, and BA. The nonzero low-order mixed measure from [5] is a required calibration; the neutral answer does not permit deleting these cross terms. Once the full joint kernel is assembled, we can return to the Susskind bulk mass/position dictionary and interaction observables. The complete de Sitter dual and finite-G four-point dynamics are not established by this microscopic neutral calculation.

All artifacts remain local in this stage. There is no Git commit, push, archive, or external issue reply.

References

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